MOLECULAR DYNAMICS SIMULATIONS OF BUBBLE NUCLEATION IN METASTABLE LENNARD-JONES FLUIDS: DETERMINING NUCLEATION THRESHOLD FACTORS

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Abstract

Bubble chambers are a leading technology for the direct detection of dark matter, sensitive to nuclear recoils from weakly interacting massive particles (WIMPs). We traditionally describe the energy threshold for bubble nucleation by the Seitz heat spike model, which combines classical nucleation theory with assumptions about localized energy deposition. Calibrations have revealed the actual nucleation thresholds to exceed Seitz predictions by factors of two or more, a result supported by some molecular dynamics (MD) simulations.

In this thesis, we present MD simulations of bubble nucleation in metastable Lennard-Jones fluids. We use two approaches. Firstly, we utilize the NVT-seeding method to determine the barriers for thermodynamic nucleation (Q_{Seitz}) by equilibrating the critical bubbles at controlled densities. The second approach involves hot spike simulations that inject localized kinetic energy to identify dynamic nucleation thresholds in NVE mode.

We performed simulations using HOOMD-blue on systems of $N = 32{,}000$ and $N = 64{,}000$ particles interacting via a truncated force-shifted Lennard-Jones potential with cutoff $r_c = 2.5\,\sigma$. At reduced density $\rho^* = 0.61$ and temperature $T^* = 0.785$, seeding simulations yield near-critical thermodynamic barriers of $Q_{\mathrm{Seitz}} \approx 170$–$440\,\varepsilon$. Hot spike simulations at $\rho^* = 0.625$ show nucleation thresholds of $KE_{\mathrm{threshold}} \approx 500\,\varepsilon$ ($N = 64{,}000$) for spherical energy deposition.

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Chapter 1

Introduction

1.1 The Dark Matter Problem

One of the mysteries of modern physics is the nature of dark matter. Observations, from the rotation curves of galaxies to the cosmic microwave background, indicate that approximately 85% of the matter in the universe interacts only weakly with electromagnetic radiation [1]. Among the leading candidates for dark matter are the Weakly Interacting Massive Particles (WIMPs), which cannot be composed of any known particle in the standard model. WIMPs are hypothetical particles with masses in the range of 1–1000 GeV/c^2 that interact with ordinary matter through the weak nuclear force. If the dark matter halo of our galaxy consists of WIMPs, they should occasionally scatter off atomic nuclei, depositing energy in the form of nuclear recoils, and direct detection experiments aim to identify these rare interactions expected at rates below one event per kilogram per year.

1.2 Bubble Chambers for Dark Matter Detection

Bubble chambers were originally developed in the 1950s by Glaser for particle physics experiments as tools for dark matter searches. A liquid is maintained in a superheated, metastable state at a temperature above its boiling point but below the spinodal limit in a bubble chamber. In a free energy state, due to the surface tension, the liquid is thermodynamically metastable with respect to vapor formation; kinetic barriers prevent spontaneous boiling. When energy is deposited in a localized region by particle interaction, it can overcome this barrier and trigger the nucleation of a macroscopic bubble.

The PICO collaboration is a modern example of using a superheated bubble chamber to search for WIMPs. They operated with 52 kg of C_3F_8 at the underground SNOLAB laboratory. For a run, the pressure in the chamber is lowered, superheating the C_3F_8 liquid and placing the detector in a live state. This metastable state is ideal for observing when a localized energy deposition can nucleate phase transitions that can lead to a macroscopic bubble [2].

1.3 Scintillating Bubble Chambers

A significant advance in the bubble chamber technology has been through the Scintillating Bubble Chamber (SBC) collaboration, with significant contributions from Professor Eric Dahl's group at Northwestern University [3]. They operated with a xenon bubble chamber and observed bubble nucleation and scintillation signals in a superheated liquid. Their operating framework follows the Seitz "hot spike" model, in which minimum energy deposition of E_T within a critical radius r_c nucleates a bubble. These quantities are determined by properties such as surface tension, vapor pressure, and the heat of vaporization. At a thermodynamic Seitz threshold of 8.3 keV, they recorded the minimum nucleation recoil energy that is required to nucleate a bubble to be 19 ± 6 keV, giving a "Seitz" factor of ≈ 2.3. They operate at thresholds where the spontaneous nucleation rate is « 1 event per ton-year

The SBC program and results highlight that bubble nucleation depends on the total deposited recoil energy as well as how that energy is localized. It also makes the Seitz threshold a baseline for microscopic studies to investigate how the deposited energy in a localized region is converted into a bubble that grows.

1.4 Thesis Objectives and Approach

The goal of this thesis is to determine Seitz factors through direct molecular dynamics (MD) simulations of bubble nucleation in metastable Lennard-Jones (LJ) Fluids. The computational approach is as follows:

NVT-Seeding simulations Following the methodology of Rosales-Pelaez et al. [4], we create critical bubbles by removing particles from spherical regions in equilibrated liquid conditions. We then allow the system to relax at a fixed volume and temperature. We determine the Seitz threshold energy, Q_{Seitz} by using the difference between the seeded and liquid energies, in addition to a correction term for the removed particles.

NVE hot spike simulations We inject localized kinetic energy into equilibrated liquid using the Maxwell- Boltzmann velocity distribution. We then evolve the system, in NVE, to determine if nucleation takes place. We extract the threshold energies from nucleation across replicate runs, and we use the spike-centered radius, $R(t)$, to quantify bubble growth. Using the thermodynamic thresholds from seeding and the hot-spike threshold, we then determine Seitz factors.

By comparing the results from these two methods, we then characterize the dynamic energy deposition needed for bubble nucleation. We use the truncated force-shifted Lennard-Jones potential [5], a similar model employed in previous nucleation studies.

Chapter 2

Theoretical Background

In this chapter, we present the theoretical framework and foundations for understanding molecular dynamics simulations of bubble nucleation. We start with the classical nucleation theory (CNT), which provides the necessary thermodynamic framework for understanding phase transitions in metastable systems. Then we discuss the Lennard-Jones potential and the truncated force-shifted Lennard-Jones that we used in our simulations. Lastly, we discuss the thermodynamics of metastable fluids.

2.1 Classical Nucleation Theory

Classical Nucleation Theory (CNT) helps in understanding the thermodynamics of bubble formation in metastable liquids.

2.1.1 Free Energy of Cluster Formation

Let us consider a metastable liquid at pressure p_l that is thermodynamically metastable with respect to the vapor formation. This liquid can exist in this state due to the kinetic barrier that is associated with creating a new liquid-vapor interface. For a vapor bubble to form, the system must compete between the energy cost required to create a liquid-vapor interface and the free energy of converting liquid to vapor. According to CNT, the free energy that is required to form a vapor bubble of radius R is given by:

$$\Delta G(R) = 4\pi R^2\gamma + \frac{4}{3}\pi R^3 \Delta g_v, \tag{2.1}$$

where γ is the liquid–vapor surface tension and Δg_v is the bulk free-energy density difference between the liquid and vapor phases. For bubble nucleation in a metastable liquid, $\Delta g_v < 0$, meaning that the vapor phase is thermodynamically favored in the bulk. For small bubbles, the surface term dominates and $\Delta G > 0$, which makes bubble formation unfavorable. As the bubble grows, ΔG decreases, and the volumetric term will eventually dominate.

2.1.2 Critical Radius and Nucleation Barrier

As mentioned in the previous chapter, the differences between the surface and the volume term produce a maximum in $\Delta G(R)$ at a critical radius R_c. Differentiating Eq. (2.1) and setting $d\Delta G/dR = 0$:

$$8\pi R\gamma + 4\pi R^2 \Delta g_v = 0 \tag{2.2}$$

which yields the critical radius:

$$R_\mathrm{c} = -\frac{2\gamma}{\Delta g_v} = \frac{2\gamma}{\Delta p} \tag{2.3}$$

where we have used $\Delta p = p_v - p_l = -\Delta g_v$ for an incompressible liquid, with p_v being the vapor pressure inside the bubble. Bubbles which are smaller than R_c tend to collapse due to surface tension (γ), whereas bubbles that are larger than R_c will grow spontaneously due to the decrease in free energy with increasing size.

We get the free energy of the critical nucleus by substituting Eq. (2.3) into Eq. (2.1):

$$\Delta G^* = \frac{16\pi\gamma^3}{3(\Delta p)^2} = \frac{4}{3}\pi R_\mathrm{c}^2 \gamma \tag{2.4}$$

Which is the maximum free-energy cost required to form a bubble.

2.2 The Lennard-Jones Potential

For the thesis, to model the metastable liquid, we use the Lennard-Jones (LJ) fluid, which is a standard fluid model commonly used in molecular dynamics studies.

2.2.1 Functional Form

The standard Lennard-Jones potential describes the interaction energy between two particles separated by distance r [6]:

$$u_{\mathrm{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right] \tag{2.5}$$

where ε is the depth of the potential well (the binding energy), and σ is the distance at which the potential equals zero.

2.2.2 Reduced Units

All simulations in this thesis are reported in reduced Lennard–Jones units, using ε, σ, and the particle mass m as the fundamental energy, length, and mass scales. The corresponding reduced quantities are

Quantity	Reduced Unit	Symbol
Length	σ	$r^* = r/\sigma$
Energy	ε	$E^* = E/\varepsilon$
Temperature	$\varepsilon/k_{\mathrm{B}}$	$T^* = k_{\mathrm{B}}T/\varepsilon$
Time	$\tau = \sigma\sqrt{m/\varepsilon}$	$t^* = t/\tau$
Density	σ^{-3}	$\rho^* = \rho\sigma^3$
Pressure	ε/σ^3	$p^* = p\sigma^3/\varepsilon$

2.2.3 Truncated Force-Shifted Potential

The LJ potential is truncated at some cutoff distance r_c to make simulations a bit less computationally intensive. The standard approach is to set $u(r) = 0$ for $r > r_c$; however, this leads to energy drift in NVE simulations.

Therefore, an alternate approach is the *truncated force-shifted* (TFS) potential. This approach ensures that both the force and the potential continuously go to zero at the cutoff:

$$u_{\mathrm{TFS}}(r) = \begin{cases} u_{\mathrm{LJ}}(r) - u_{\mathrm{LJ}}(r_c) - (r - r_c) \left.\dfrac{du_{\mathrm{LJ}}}{dr}\right|_{r=r_c}, & r \leq r_c \\ 0, & r > r_c \end{cases} \tag{2.6}$$

with corresponding force

$$f_{\mathrm{TFS}}(r) = \begin{cases} f_{\mathrm{LJ}}(r) - f_{\mathrm{LJ}}(r_c), & r \leq r_c \\ 0, & r > r_c. \end{cases} \tag{2.7}$$

In this thesis, we use a cutoff of $r_c = 2.5\sigma$.

2.3 Thermodynamics of Metastable Fluids

2.3.1 Phase Diagram and Coexistence

Below the critical temperature, the LJ fluid consists of coexisting liquid and vapor phases at equilibrium. Along the coexistence curve, both phases should satisfy these conditions:

$$T_l = T_v, \tag{2.8}$$

$$p_l = p_v, \tag{2.9}$$

$$\mu_l = \mu_v. \tag{2.10}$$

Bubble nucleation takes place when a liquid is prepared away from its coexistence condition in a metastable state, which is the point of focus in out thesis.

2.3.2 Metastability: Stretched and Superheated Liquids

The liquid is metastable with respect to the vapor formation when it is either stretched below its equilibrium vapor pressure (at fixed T) or when it is superheated above its boiling temperature (at fixed pressure).

There have been studies that considered stretched liquids using NVT seeding, as well as those that examine superheated liquids using NPT [4, 7]. Our thesis focuses on metastable LJ fluids to analyze bubble formation conditions through seeding and hot-spike simulations.

2.4 The Seitz Heat Spike Model

The Seitz "hot-spike" model provides a framework for bubble nucleation in a superheated liquid [8]. Bubble formation occurs when energy is deposited in a localized region. For our work, the Seitz hot-spike model serves as a useful baseline, as, in a microscopic energy deposition, not all of the deposited energy will remain localized within the volume. Some of that energy is lost through energy diffusion and non-equilibrium processes before a stable bubble can form. In our simulations, we explore NVT seeding to determine the classical Seitz "hot-spike" thermodynamic threshold, and NVE hot-spike simulations to determine the dynamic energy required for successful bubble nucleation.

Chapter 3

Simulation Methodology

In this chapter, we describe the computational methods used to study the bubble nucleation in metastable LJ fluids. We use two approaches: NVT-seeding for determining nucleation barriers, and NVE hot spike simulations for determining Seitz factors. Both methods are implemented using the HOOMD-blue molecular dynamics package running on GPUs.

3.1 Simulation Setup

All simulations in this thesis were performed with HOOMD-blue v4.x using GPU acceleration. Interparticle interactions were modeled with the truncated force-shifted Lennard–Jones (TFS-LJ) potential introduced in Chapter 2, using a cutoff $r_c = 2.5\,\sigma$. This choice was made to improve force continuity at the cutoff and to reduce energy drift during the long NVE trajectories used in the hot-spike calculations.

3.1.1 Initial Liquid Configuration

The simulation cell is a cubic box with periodic boundary conditions in all three directions. For a target reduced density ρ, the box length is

$$L = \left(\frac{N}{\rho}\right)^{1/3}. \tag{3.1}$$

The primary simulations use $N = 32{,}000$ particles. We also carried out finite-size effect checks with $N = 64{,}000$.

Initial particle positions are generated on a cubic grid with spacing L/n_{side}, where $n_{\text{side}} = \lceil N^{1/3} \rceil$, and each coordinate is then Given a small random offset to break the exact lattice symmetry. This provides a convenient starting configuration that rapidly relaxes to a homogeneous liquid during the subsequent equilibration stage.

3.1.2 Equilibration

Initial particle velocities are initialized using a Maxwell-Boltzmann velocity distribution at the target reduced temperature $T = 0.785$. The system is then equilibrated in the NVT ensemble using the MTTK thermostat in HOOMD-blue, with timestep

$$\Delta t = 0.0012\,\tau \tag{3.2}$$

and thermostat coupling time

$$\tau_T = 0.46\,\tau. \tag{3.3}$$

This equilibrated liquid is the starting point for both the simulations used in the thesis. In the seeding simulations, we modify the equilibrated liquid by creating a cavity and then evolving further in NVT. In the hot-spike simulations, we first relax the equilibrated liquid in NVE before we apply localized energy injection.

3.1.3 Simulation Parameters

The key simulation parameters are summarized in Table 3.1.

Table 3.1: Simulation parameters used in this work (reduced Lennard–Jones units).

Parameter	Symbol	Value
Number of particles	N	32,000 (primary); 64,000 (finite-size check)
Reduced temperature	T^*	0.785
Reduced density (seeding)	ρ^*	0.61
Intended seed radii	R_i	$5.7,\ 6.8,\ 7.6,\ 8.5,\ 9.4,\ 10.3,\ 11.2,\ 12.2\,\sigma$
Reduced density (hot spike)	ρ^*	0.625 (primary)
Pair potential cutoff	r_c	$2.5\,\sigma$
Integrator timestep (normal)	Δt	$0.0012\,\tau$
Integrator timestep (post-spike)	Δt_{spike}	$0.0001\,\tau$
Thermostat (NVT)	—	MTTK thermostat
Thermostat coupling	τ_T	$0.46\,\tau$
NVT equilibration steps	N_{eq}	10^5
Pre-spike NVE steps	N_{pre}	10^4
Post-spike fine-step steps	N_{fine}	5×10^4
Post-spike observation steps	N_{obs}	10^5
Replicates per energy (threshold scans)	N_{rep}	10

3.2 NVT-Seeding Methodology

The seeding simulations in the thesis follow the strategy for fixed-volume bubble nucleation of Rosales-Pelaez et al. [4]. In the NVT ensemble, with fixed-volume, the formation of a vapor cavity slightly compresses the surrounding liquid. This state allows a critical bubble to be in near equilibrium, making it possible for us to measure its properties.

3.2.1 Seeded-Bubble Preparation and Equilibration

For each intended seed radius R_i, we begin from an equilibrated homogeneous liquid, then a spherical cavity is created by removing all particles whose positions lie within a sphere of radius R_i centered at the box origin. This results in a low-density region that serves as the initial bubble seed.

After we remove particles in a given region, the simulation continues to evolve in NVT. Although we do not utilize a spherical repulsive wall model as Rosales's group does, we hold the cavity briefly centered at the seed location. The seeded configuration is then further equilibrated for 2×10^5 steps, during which we monitor the bubble radius from the radial density profile. A production run of 10^5 additional steps is then performed, with potential energy and pressure sampled every 10^3 steps.

We extract the bubble properties from the radial density profiles, which we compute around the bubble center and fit to the hyperbolic-tangent form

$$\rho(r) = \frac{\rho_v + \rho_l}{2} + \frac{\rho_l - \rho_v}{2} \tanh\left(\frac{r - R_c}{\alpha}\right), \tag{3.4}$$

where ρ_v and ρ_l are the fitted vapor and liquid densities, R_c is the fitted midpoint radius, and α is the interfacial width.

In the results, we report the fitted equilibrium bubble radius as R_f.

3.2.2 Corrected Fixed-Volume Threshold Energy

To characterize the thermodynamic cost of bubble formation in the seeded runs, we evaluate a corrected fixed-volume threshold energy, denoted here by Q_{Seitz}. Let (N_0, U_0, P_0) denote the particle number, total potential energy, and pressure of a homogeneous metastable liquid reference state, and let (N_c, U_c) denote the corresponding particle number and total potential energy for the equilibrated seeded state containing the bubble. Here, the reference state (N_0, U_0, P_0), and the seeded state, (N_c, U_c), must be compared only at the same box volume V and the same surrounding liquid density. With $\Delta N = N_0 - N_c$, we use

$$Q_{\text{Seitz}} = (U_c - U_0) + \frac{\Delta N}{N_0} (U_0 + P_0 V) . \tag{3.5}$$

The first term is the raw total potential-energy difference between the seeded and homogeneous systems, while the second term accounts for the fixed-volume thermodynamic work associated with the removed particles.

3.3 NVE Hot Spike Methodology

As mentioned in the earlier chapters, for the hot-spike simulations, we model localized energy deposition in a metastable liquid and check whether that deposition energy produces a bubble. For each replicate, we prepare one equilibrated liquid configuration at the target (N, V, T) by running an NVT equilibration followed by a short NVE relaxation. We re-use this pre-spike state for all the energy injections at that (N, V, T) and compare the growth of the bubbles resulting from the different injected energies on the same underlying liquid configuration.

3.3.1 Localized Thermalized Injection and NVE Evolution

We implement the energy deposition inside a spherical injection region of radius R_{inj} centered at the origin. For the work presented here, we use

$$R_{\text{inj}} = 3.0\,\sigma. \tag{3.6}$$

For a given pre-spike state prepared, all particles with distance $r < R_{\text{inj}}$ from the injection centers are identified, and their velocities are re-thermalized using a Maxwell-Boltzmann velocity distribution at an elevated spike temperature T_{spike}.

For each injection radius, the corresponding kinetic-energy is

$$KE^{(\text{req})}_{\text{inject}} = \frac{3}{2} N_{\text{inj}} k_B \left(T_{\text{spike}} - T_0\right), \tag{3.7}$$

where N_{inj} is the number of particles in the heated region and T_0 is the pre-spike kinetic temperature of that region.

After we re-thermalize, the system then evolves in NVE, and the change in total energy,

$$\Delta E_{\text{total}} = E_{\text{after inject}} - E_{\text{before}}, \tag{3.8}$$

is measured from the simulation directly.

After the energy injection, we integrate the system using a reduced timestep $\Delta t_{\text{spike}} = 0.0001\,\tau$ for 5×10^4 steps to resolve the rapid initial dynamics. The timestep is then returned to its normal value, $\Delta t = 0.0012\,\tau$, and we continue the trajectory for an additional 10^5 steps. We monitor bubble growth throughout the NVE evolution using the radial profile.

3.3.2 Threshold Definition

For each of the initial requested injection energy, we evaluate the nucleation response over a set of 10 replicates. For any given replicate, we use the same pre-spike state prepared for all the energies tested.

Furthermore, to avoid any false positives, we examine the pre-spike state using the same density profile method used after injection. If the pre-spike bubble radius, which normally should be zero, at the spike center exceeds a threshold, we exclude the run from the threshold count. The number of valid replicates at fixed (T, ρ) serves as a diagnostic for whether the points chosen are suitable for the threshold measurement.

Then, for each of the valid trajectories, we determine two metrics from the spike-centered bubble radius:

$$R_{\max} \equiv \max_t R(t), \tag{3.9}$$

$$R_{\text{late}} \equiv \langle R(t) \rangle_{\text{late}}, \tag{3.10}$$

where $R_{\max}$ measures the largest transient growth reached during the trajectory and R_{late} is the late-time average radius over the final portion of the run.

At each of the requested energies, we also compute the nucleation probability as the fraction of valid replicates that satisfy the event criterion. In Chapter 5 we report transient thresholds based on $R_{\max} \geq 3\sigma$, 4σ, and 5σ, together with a late-time stability criterion based on R_{late}. The threshold is defined as the smallest requested energy for which the corresponding probability satisfies

$$P \geq 0.5. \tag{3.11}$$

3.4 Bubble Detection and Analysis

In both the seeding and hot-spike simulations, we use a radial density profile for bubble detection, together with a smooth interfacial fit. For the hot-spike simulations, the profile is always centered at the injection point, so we only count the responses at the location of energy deposition, whereas for the seeding simulations, the bubble may drift; hence, we first use a coarse low-density profile to estimate the bubble center before constructing the full radial profile.

3.4.1 Radial Density Profile and Interface Fit

For a given valid bubble center, we compute the number density $\rho(r)$ in spherical shells using periodic boundary conditions. We then fit the resulting profile to the hyperbolic-tangent form:

$$\rho(r) = \frac{\rho_v + \rho_l}{2} + \frac{\rho_l - \rho_v}{2} \tanh\left(\frac{r - R_c}{\alpha}\right), \tag{3.12}$$

where ρ_v and ρ_l are the fitted vapor and liquid densities, R_c is the midpoint radius, and α is the interfacial width.

For the hot-spike calculations and analysis, we classify a frame as "bubbled" only if the fitted profile and the density estimates all satisfy these conditions:

- the density inside a core region of radius 3σ satisfies

$$\rho_{\text{core}} \leq 0.5\,\rho^*; \tag{3.13}$$

- the far-field density estimated from the outer part of the profile satisfies

$$\rho_{\mathrm{bulk}} \geq 0.85\, \rho^*; \tag{3.14}$$

- the fitted vapor density satisfies

$$\rho_v \leq 0.2\, \rho_l; \tag{3.15}$$

- the fitted midpoint radius satisfies

$$R_c \geq 3\sigma; \tag{3.16}$$

- the fitted interface width satisfies

$$\alpha \leq 0.5\, R_c. \tag{3.17}$$

If for a given candidate, any one of these conditions fails, or if the fit fails, we classify that frame as non-bubbled and set the reported radius to zero.

Chapter 4

NVT Seeding Simulation Results

In this chapter, we present the results of the NVT seeding simulations as described in Chapter 3. The initial state points chosen are $\rho_l^* = 0.61$ at $T^* = 0.785$, for direct comparison with the NVT-Seeding study of Rosales-Pelaez *et al.* [4]. We then present the extracted bubble radii, density profiles, and liquid pressures from our HOOMD-blue code implementation. We then compare against the LAMMPS-based results of Rosales-Pelaez *et al.*. We also report the corrected fixed-volume threshold quantity, Q_{Seitz}, and provide a discussion regarding the referenced liquid state used in the calculation.

4.1 Simulation Details

The seeding calculations reported here use $N_0 = 32{,}000$ particles in a cubic periodic box at $T^* = 0.785$ and initial liquid density $\rho_i^* = 0.61$. Eight intended cavity radii were studied, $R_i = 5.7,\ 6.8,\ 7.6,\ 8.5,\ 9.4,\ 10.3,\ 11.2$, and $12.2\,\sigma$, similar to the set used by Rosales-Pelaez *et al.* [4].

4.2 Results

4.2.1 Summary of Seeding Runs

Table 4.1 summarizes the results of eight seeding simulations spanning intended radii from $R_i = 5.7\,\sigma$ to $12.2\,\sigma$. The same set of R_i values was used by Rosales-Pelaez *et al.*, which allows for a direct comparison of the bubble properties.

Table 4.1: NVT seeding results at $\rho_l^* = 0.61$ and $T^* = 0.785$. R_i is the intended cavity radius, R_f, the fitted equilibrium radius from the sigmoid density profile, $\rho_{v,f}$ and $\rho_{l,f}$ the fitted vapor and liquid densities, $P_{v,f}$ and $P_{l,f}$ the vapor and liquid pressures, ΔN the number of particles removed, U_0 and U_c the potential energies of the homogeneous and seeded systems, $\Delta\text{PE}_{\text{raw}} = U_c - U_0$, and Q_{Seitz} the corrected threshold energy from Eq. (3.5). All quantities are in LJ reduced units.

R_i	R_f	$\rho_{v,f}$	$\rho_{l,f}$	$P_{v,f}$	$P_{l,f}$	N_0	N_c	ΔN	U_0	U_c	$\Delta\text{PE}_{\text{raw}}$	Q_{Seitz}
5.7	9.21	0.0245	0.6605	0.0192	−0.0165	32000	31465	535	−121423.50	−118938.33	2485.17	437.63
6.8	9.59	0.0326	0.6633	0.0256	−0.0090	32000	31158	842	−121215.17	−117809.27	3405.90	184.79
7.6	9.88	0.0380	0.6721	0.0298	−0.0079	32000	30786	1214	−121074.52	−116268.94	4805.59	168.68
8.5	8.79	0.0287	0.5994	0.0225	−0.0029	32000	30332	1668	−121099.75	−114368.57	6731.18	362.93
9.4	10.15	0.0280	0.6598	0.0220	−0.0095	32000	29792	2208	−121463.43	−111698.62	9764.81	1334.43
10.3	9.59	0.0290	0.5800	0.0228	0.0026	32000	29011	2989	−121308.79	−108695.11	12613.69	1192.42
11.2	9.86	0.0267	0.5640	0.0210	0.0007	32000	28229	3771	−120815.03	−105219.50	15595.53	1200.38
12.2	11.52	0.0298	0.6005	0.0234	−0.0033	32000	27029	4971	−121353.87	−99429.85	21924.02	2943.48

The fitted vapor densities are uniformly low ($\rho_{v,f} \approx 0.02$–0.04), and he raw potential-energy difference, $\Delta\text{PE}_{\text{raw}} = U_c - U_0$, increases monotonically with The number of removed particles is expected for larger seeded cavities. However, the fitted liquid densities and bubble radii show a non-monotonic trend with the seed radius, R_i. This behavior can be interpreted in regards to the reference liquid state used in the calculation, which does not match the equilibrated far-field density $\rho_{l,f}$ of each seeded configuration.

The Q_{Seitz} values listed were evaluated with the homogeneous reference state at the initial liquid density $\rho_i^* = 0.61$. However, there is a discrepancy, where the corrected fixed-volume comparison should instead use a pure-liquid reference at the same box volume and the same surrounding liquid density as the equilibrated seeded state, i.e. using the Q_{Seitz} and fitted far-field liquid density $\rho_{l,f}$. This is the reason for the non-monotonic behavior between our results and their work.

We also compare the seeded states more directly with the results of Rosales-Pelaez *et al.* In the figure below, we plot the fitted far-field liquid density against the fitted equilibrium bubble radius for both datasets. This comparison is useful, as discussed before, that the relevant quantities for the equilibrated seeded states are the R_f and $\rho_{l,f}$, rather than the initial seed parameters alone.

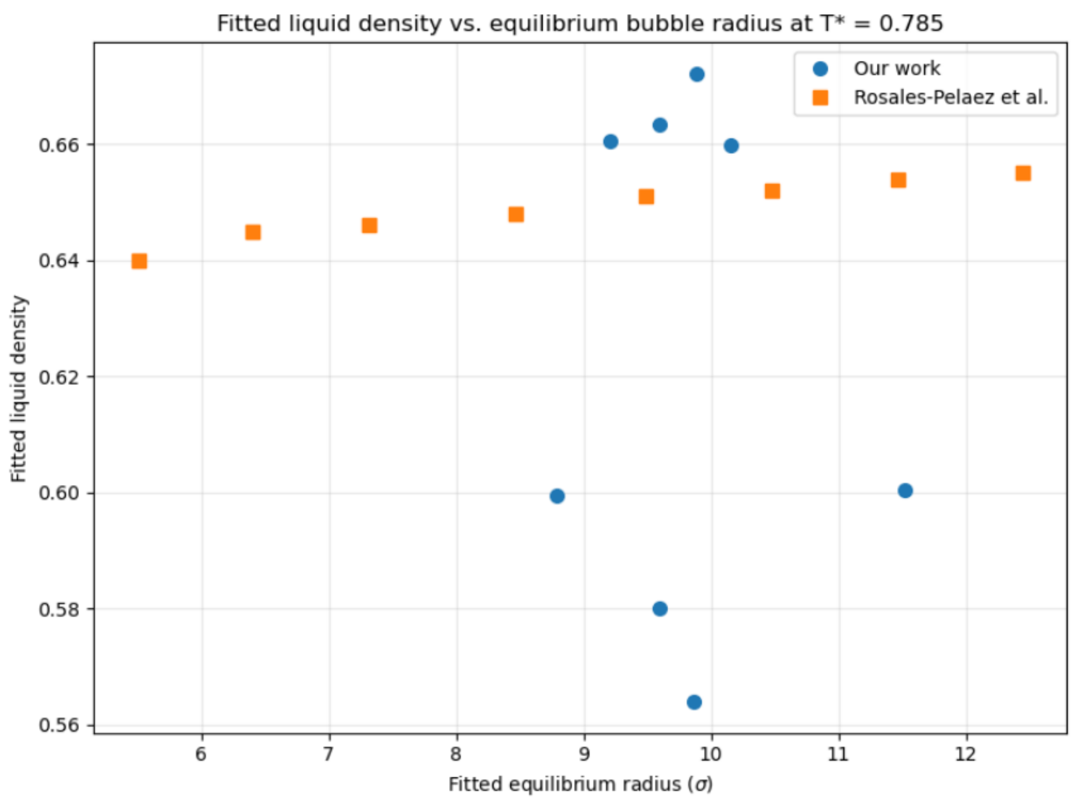

Figure 4.1: Comparison of fitted far-field liquid density versus fitted equilibrium bubble radius for our seeding runs and those of Rosales-Pelaez *et al.* [4]. It is noticeable that the Rosales data follows a relatively smooth trend, remaining in a narrow range as the fitted bubble radius increases. However, our result shows a much broader scatter. This is consistent with our understanding that some of the seeded runs may not yet be fully equilibrated, and finite-size effects also become important once the bubble occupies a good fraction of the box.

4.2.2 Radial Density Profiles

We also verify that the sigmoid fit provides an accurate description of the equilibrated bubble structure. We use Figure 4.2 to show the radial density profiles for three seeded radii spanning the lower, middle, and upper portions of the intended seeded-radii range. In each case, the density histogram is smoothly captured by the sigmoid fit, and the fitted bubble radius does lie near the midpoint of the transition region.

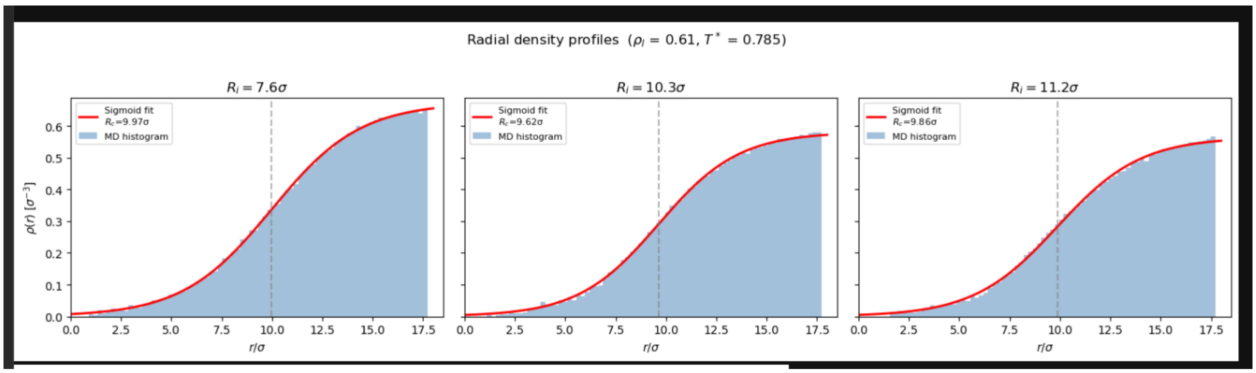

Figure 4.2: Radial density profiles at $\rho_l^* = 0.61$ and $T^* = 0.785$ for seeded radii $R_i = 7.6\,\sigma$, $10.3\,\sigma$, and $11.2\,\sigma$. The shaded histograms show the binned data, the solid curves are the sigmoid fits, and the vertical dashed lines denote the fitted equilibrium radius R_f. There is a close agreement between the histograms and the fits, which confirms that the bubble radius can be extracted consistently from the late-time density profiles. Note that the end of the plot is where the diameter equals the box size.

4.2.3 Fitted Bubble Radius

The fitted equilibrium radii reported in Table 4.1 cluster in a relatively narrow range, $R_f \approx 8.8$–$11.5\,\sigma$, even though the intended seed radii span $R_i = 5.7$–$12.2\,\sigma$. The smaller seeds grow during equilibration, and the largest seed contracts, which indicates that the seeded systems relax towards a range of late-time bubble sizes in the fixed-volume configuration.

Figure 4.3 shows the fitted radius as a function of time for three representative seeds. The $R_i = 7.6\,\sigma$ seed grows toward a larger late-time radius, while the $R_i = 12.2\,\sigma$ seed contracts. The $R_i = 10.3\,\sigma$ case, however, still shows a downward trend at the end of the production window, suggesting that at least some seeded runs may not yet be fully equilibrated. This is important as when comparing with Rosales-Pelaez *et al.*, whose final radii spans a much broader range. Rather than a physical difference between the two approaches, this behavior indicates that our seeding method requires much longer equilibration times.

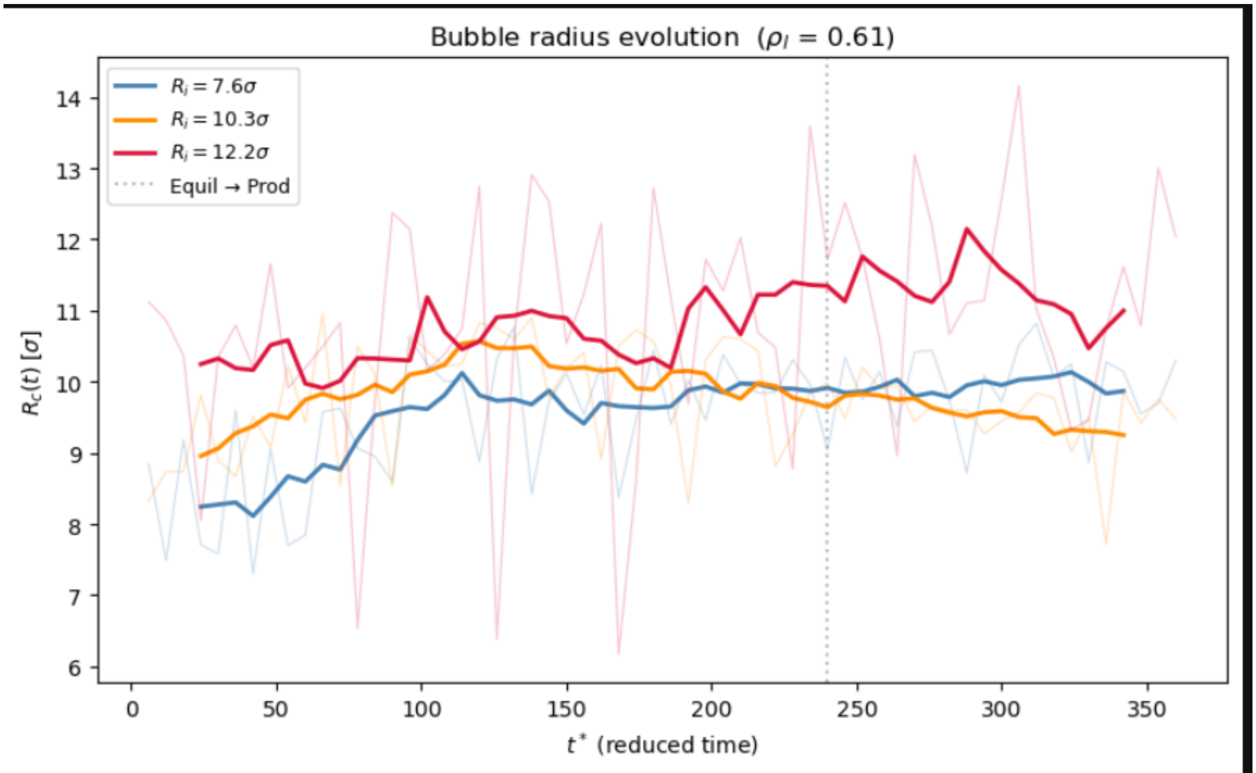

Figure 4.3: Bubble radius evolution curves at $\rho_l^* = 0.61$. The $R_i = 7.6\,\sigma$ case grows, the $R_i = 12.2\,\sigma$ case contracts, and the $R_i = 10.3\,\sigma$ case continues to drift downward late in the run, indicating that this configuration may not yet be fully equilibrated. The dotted vertical line marks the approximate transition from the equilibration stage to the production stage.

4.2.4 Liquid Density and Pressure

The fitted far-field liquid densities and the liquid pressures indicate that for smaller seeds, the fitted liquid density remains near $\rho_{l,f} \sim 0.66$–0.67 and the fitted liquid pressure increases from approximately -0.017 to -0.003 as the seed size increases. However, for the larger seeds, the fitted liquid density drops to $\rho_{l,f} \sim 0.56$–0.60 and the liquid pressure fluctuates near zero.

Compared to Rosales-Pelaez *et al.*, our trends are less uniform. In their work, the fitted liquid density remained in a much narrower range, and there was a smooth increase in liquid pressure with the bubble size. In our runs, the broader scatter does reflect finite=size effects and equilibration times, as when the remaining liquid shell becomes thin, the far-field liquid region is less well defined. This leads the fitted liquid density to be much more sensitive to the box size and the incomplete relaxation period.

4.2.5 Corrected Seitz Threshold Energy

The Q_{Seitz}, we obtained is positive for all the seeded configurations at $\rho_i^* = 0.61$. They range from approximately $170\,\varepsilon$ to $2950\,\varepsilon$, and there is a non-monotonic behavior across the eight seeded runs.

These values should be interpreted with the earlier statement that the corrected fixed-volume comparison is formally intended to use a homogeneous liquid reference state at the same box volume and the same surrounding liquid density as the equilibrated seeded configuration. In the present analysis, however, the homogeneous reference was taken from the initial liquid state at $\rho_i^* = 0.61$ rather than from a pure-liquid state matched to the fitted far-field density $\rho_{l,f}$.

4.3 Comparison with Rosales-Pelaez *et al.*

In Table 4.2, we compare the fitted quantities from our simulations with the corresponding values from Table I of Rosales-Pelaez *et al.* Both datasets use the same intended radii R_i, similar particle count $N_0 = 32{,}000$, and comparable box sizes.

Table 4.2: Comparison of fitted bubble properties at $\rho_l^* = 0.61$ between our work (HOOMD-blue) and Rosales-Pelaez *et al.* [4] (LAMMPS). R_f and R_c denote the fitted equilibrium radius from each study, $\rho_{l,f}$ and $\rho_{l,\mathrm{dp}}$ the fitted liquid densities, and $P_{l,f}$ and $P_{l,\mathrm{dp}}$ the liquid pressures.

	Fitted Radius		Liquid Density		Liquid Pressure	
R_i	R_f	$R_c^{\mathrm{Ros.}}$	$\rho_{l,f}$	$\rho_{l,\mathrm{dp}}^{\mathrm{Ros.}}$	$P_{l,f}$	$P_{l,\mathrm{dp}}^{\mathrm{Ros.}}$
5.7	9.21	6.40	0.661	0.645	−0.017	−0.031
6.8	9.59	5.51	0.663	0.640	−0.009	−0.038
7.6	9.88	7.32	0.672	0.646	−0.008	−0.025
8.5	8.79	8.47	0.599	0.648	−0.003	−0.019
9.4	10.15	9.48	0.660	0.651	−0.010	−0.014
10.3	9.59	10.48	0.580	0.652	0.003	−0.011
11.2	9.86	11.47	0.564	0.654	0.001	−0.008
12.2	11.52	12.44	0.601	0.655	−0.003	−0.005

There is a point of agreement in that both studies produce well-defined equilibrated bubble profiles with low vapor densities and a surrounding liquid phase that remains in a metastable state. At the same time, the fitted late-time properties differ from those reported by Rosales-Pelaez *et al.*. Our fitted bubble radii cluster within a narrow range and our liquid density and pressure show more scatter, especially for the larger seeds. This leads to the conclusion that some of the present seeded runs may not be fully equilibrated.

From this comparison, we also conclude that finite-size effects become increasingly important for the larger seeds, as once a bubble occupies a substantial fraction of the box volume, the remaining liquid shell becomes thinner and the fitted far-field liquid density and pressure are more sensitive to limited box size and relaxation time. Despite these differences, the present results do reproduce the main qualitative feature of the NVT-seeding method, which is to observe bubble configuration and smooth liquid-vapor interfaces and analyze the trajectories in a fixed-volume configuration.

Chapter 5

Hot Spike Simulation Results

In this chapter, we present the results of the NVE hot-spike simulations. We deposit energy in a localized region into an equilibrated metastable Lennard-Jones fluid and track the subsequent bubble response under NVE evolution. The primary hot-spike state point studied here is $\rho^* = 0.625$ and $T^* = 0.785$.

The choice of $\rho^* = 0.625$ rather than $\rho^* = 0.61$, as in the seeding simulations, was made to work in a metastable liquid region that is less susceptible to spontaneous cavitation. The case of $\rho^* = 0.61$ is a stretched one. Additionally, the seeding analysis at $\rho^* = 0.625$ produced negative corrected Q_{Seitz} values. Therefore, we present the hot-spike analysis independently at this density. The main observations in this chapter are the measure injection energy, the maximum bubble radius $R_{\max}$, the late-time radius R_{late}, and the nucleation probability as functions of the requested injection energy.

We carried out the primary scan for the maximum bubble radius $R_{\max}$, the late-time radius R_{late}, and the nucleation probability as functions of the requested injection energy. For each system size, we scanned the injection energy from $300\,\varepsilon$ to $1000\,\varepsilon$ with 10 independent replicates at each energy.

5.1 Injection Calibration

Before the nucleation criteria, we verify that the thermalized injection method deposits the intended amount of kinetic energy. Since the hot-spike re-thermalizes the velocities of particles inside the injection region, we must measure the actual change in total energy from the simulation.

Here, we compare the mean measured energy change $\langle \Delta E \rangle$ with the requested injection energy for both system sizes. In both scans, the measured energy follows the overall trend well, which confirms that the thermalized injection procedure produces a controlled sequence of events. The trend is not perfect, as there is a notable deviation for the $N = 32{,}000$ run at requested $600\,\varepsilon$, where the measured mean energy falls below the neighboring points.

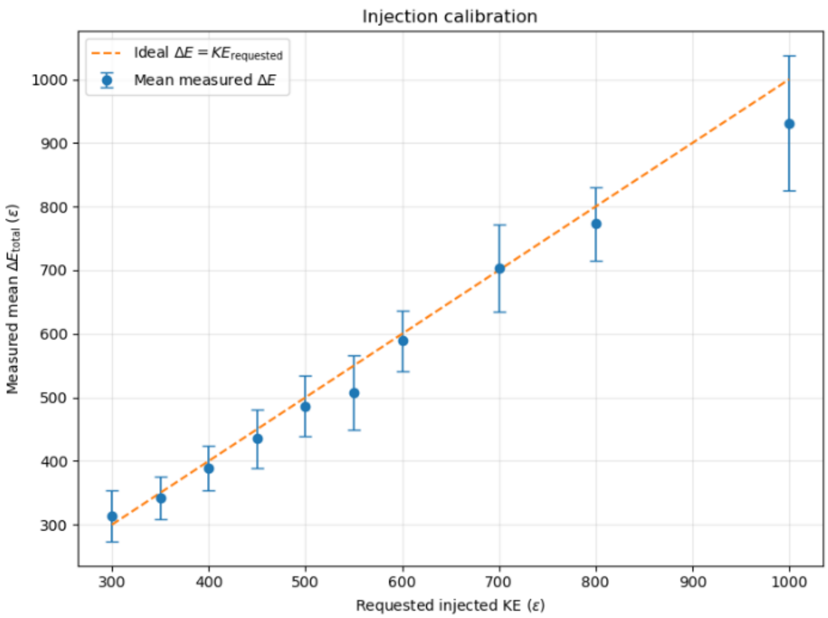

Figure 5.1: Injection calibration for the thermalized hot-spike method at $N = 32{,}000$. Points show the mean measured energy change $\langle \Delta E \rangle$ over 10 replicates at each requested injection energy, with error bars showing the standard deviation of the replicate distribution. The dashed line indicates the ideal $\Delta E = KE_{\mathrm{requested}}$ relation.

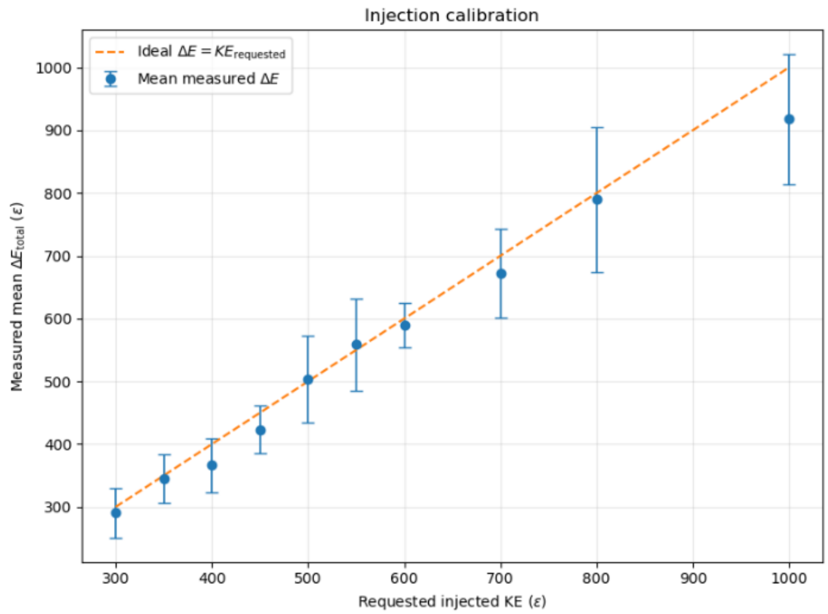

Figure 5.2: Injection calibration for the thermalized hot-spike method at $N = 64{,}000$. Points show the mean measured energy change $\langle \Delta E \rangle$ over 10 replicates at each requested injection energy, with error bars showing the standard deviation of the replicate distribution. The dashed line indicates the ideal $\Delta E = KE_{\text{requested}}$ relation.

5.2 Energy Threshold Determination

Tables 5.1 and 5.2 present the complete scan results for both system sizes. For each injected energy level, we report the number of valid replicates, the mean maximum bubble radius $\langle R_{\text{max}} \rangle$, and the mean late-time radius $\langle R_{\text{late}} \rangle$, the mean measured energy change $\langle \Delta E \rangle$, and the nucleation probabilities under each criterion. Note that at each injection energy, we performed 10 hot spike replicates. All trials successfully equilibrated to a homogeneous liquid state before heat injection, so no runs were excluded from the threshold analysis.

Table 5.1: Hot-spike scan results for $N = 32{,}000$, $\rho^* = 0.625$, and $T^* = 0.785$. Each row corresponds to 10 independent replicates at The indicated requested injection energy. $P_{R\geq n\sigma}$ denotes the fraction of replicates with $R_{\text{max}} \geq n\sigma$, and P_{stable} denotes the fraction satisfying the late-time stability criterion.

KE	$\langle R_{\text{max}} \rangle$	$\langle R_{\text{late}} \rangle$	$\langle \Delta E \rangle$	$P_{R\geq 3\sigma}$	$P_{R\geq 4\sigma}$	$P_{R\geq 5\sigma}$	P_{stable}
300	0.00	0.00	271.78	0.000	0.000	0.000	0.000
350	0.76	0.00	321.10	0.200	0.000	0.000	0.000
400	0.66	0.00	401.27	0.200	0.000	0.000	0.000
450	0.78	0.00	450.95	0.200	0.100	0.000	0.000
500	1.56	0.00	483.27	0.400	0.200	0.000	0.000
550	1.61	0.00	563.86	0.400	0.200	0.000	0.000
600	2.96	0.00	508.32	0.700	0.600	0.100	0.000
700	3.09	0.00	705.71	0.700	0.600	0.000	0.000
800	4.46	0.00	806.32	1.000	0.700	0.200	0.000
1000	4.85	0.00	1015.11	0.900	0.900	0.800	0.000

Table 5.2: Hot-spike scan results for N = 64,000, ρ^* = 0.625, and T^* = 0.785. Format identical to Table 5.1.

KE	$\langle R_{\mathrm{max}} \rangle$	$\langle R_{\mathrm{late}} \rangle$	$\langle \Delta E \rangle$	$P_{R \geq 3\sigma}$	$P_{R \geq 4\sigma}$	$P_{R \geq 5\sigma}$	P_{stable}
300	0.34	0.00	302.85	0.100	0.000	0.000	0.000
350	1.53	0.00	347.48	0.400	0.100	0.000	0.000
400	1.06	0.00	394.12	0.300	0.000	0.000	0.000
450	0.82	0.00	416.28	0.200	0.100	0.000	0.000
500	3.02	0.00	492.27	0.700	0.600	0.000	0.000
550	3.53	0.21	552.34	0.800	0.500	0.300	0.000
600	3.26	0.00	583.30	0.700	0.700	0.200	0.000
700	4.19	0.60	682.28	0.800	0.600	0.500	0.100
800	5.00	0.53	802.68	0.900	0.900	0.800	0.000
1000	5.84	0.91	965.20	1.000	1.000	1.000	0.200

The nucleation threshold depends on the event criterion used to define a successful hot-spike response. We define the threshold energy as the lowest requested injection energy for which the nucleation probability reaches or exceeds 50%. The threshold estimates are:

Table 5.3: Nucleation threshold estimates (in ε) for both system sizes under each criterion, defined as the lowest requested KE at which The nucleation probability reaches 50%.

Criterion	$N = 32{,}000$	$N = 64{,}000$
$R_{\max} \geq 3\sigma$	600	500
$R_{\max} \geq 4\sigma$	600	500
$R_{\max} \geq 5\sigma$	1000	700
Late-time $R \geq 3\sigma$	—	—

For the $N = 32{,}000$ system, the transient criteria $R_{\max} \geq 3\sigma$ and $R_{\max} \geq 4\sigma$ both yield a threshold of $600\,\varepsilon$, while the $R_{\max} \geq 5\sigma$ criterion gives a threshold of $1000\,\varepsilon$. For the $N = 64{,}000$ finite-size check, the thresholds are lower, with $500\,\varepsilon$ for the $R_{\max} \geq 3\sigma$ and 4σ criteria and $700\,\varepsilon$ for $R_{\max} \geq 5\sigma$. In contrast, the late-time criterion does not produce a 50% threshold in either system, indicating that even the 64k box is still too small to accommodate bubble growth.

5.3 Effect of Injection Parameters

Here, we compare the nucleation probability as a function of the injected energy for the four event criteria. For the late time criteria, nucleation probability still remains below 50% throughout the scan. A reasonable next step to reduce finite-size effects would therefore be to increase the system size by at least another factor of 4–8 in particle number, i.e. to $N \sim 2.5 \times 10^5$–5×10^5 particles. At $\rho^* = 0.625$, this would correspond to box lengths of approximately $L \sim 74$–$93\,\sigma$, roughly doubling the linear system size relative to the 64k runs.

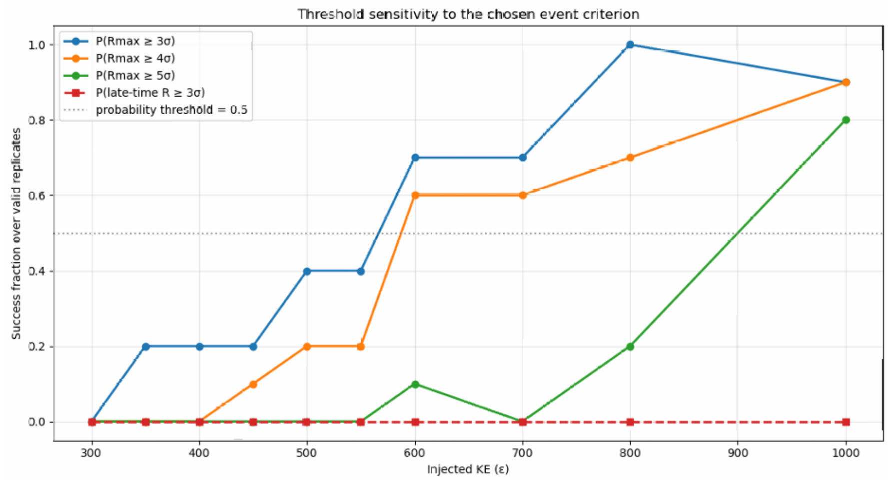

Figure 5.3: Nucleation probability as a function of the injected kinetic energy for $N = 32{,}000$ at $\rho_l^* = 0.625$. As mentioned, we show four criteria: $P(R_{\max} \geq 3\sigma)$ (blue), $P(R_{\max} \geq 4\sigma)$ (orange), $P(R_{\max} \geq 5\sigma)$ (green), and $P(\text{late-time } R \geq 3\sigma)$ (red dashed). Note that the horizontal dotted line marks the 50% probability level that we use to define the threshold energy.

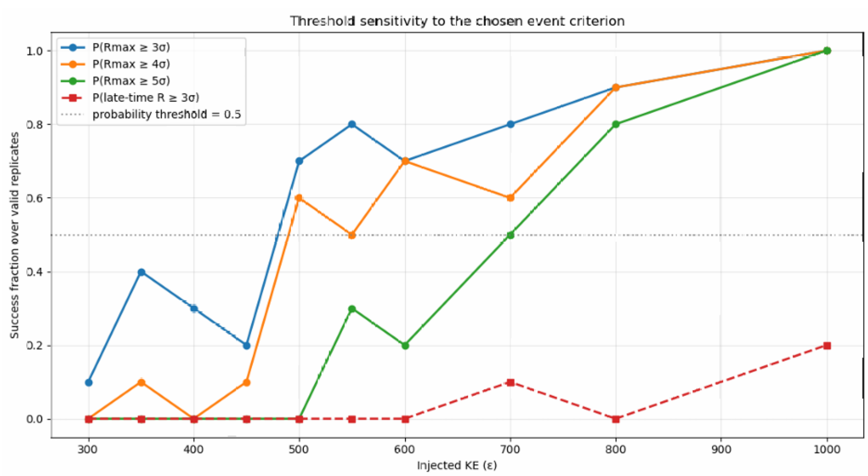

Figure 5.4: Nucleation probability as a function of the injected kinetic energy for $N = 64{,}000$ at $\rho_l^* = 0.625$. As mentioned, we show four criteria: $P(R_{\max} \geq 3\sigma)$ (blue), $P(R_{\max} \geq 4\sigma)$ (orange), $P(R_{\max} \geq 5\sigma)$ (green), and $P(\text{late-time } R \geq 3\sigma)$ (red dashed). Note that the horizontal dotted line marks the 50% probability level that we use to define the threshold energy.

we adopt the $R_{\max} \geq 4\sigma$ criterion as the primary definition of the hot-spike threshold, which gives:

$$KE_{\text{threshold}} \approx \begin{cases} 600\,\varepsilon, & N = 32{,}000, \\ 500\,\varepsilon, & N = 64{,}000. \end{cases} \tag{5.1}$$

5.4 $R(t)$ Evolution and Nucleation Criteria

The time evolution of the bubble radius provides the most direct visualization of the nucleation dynamics. Figure 5.5 shows $R(t)$ for the $N = 32{,}000$ system across the full range of injected energies. Each curve is essentially one replicate, with different colors indicating the injected KE.

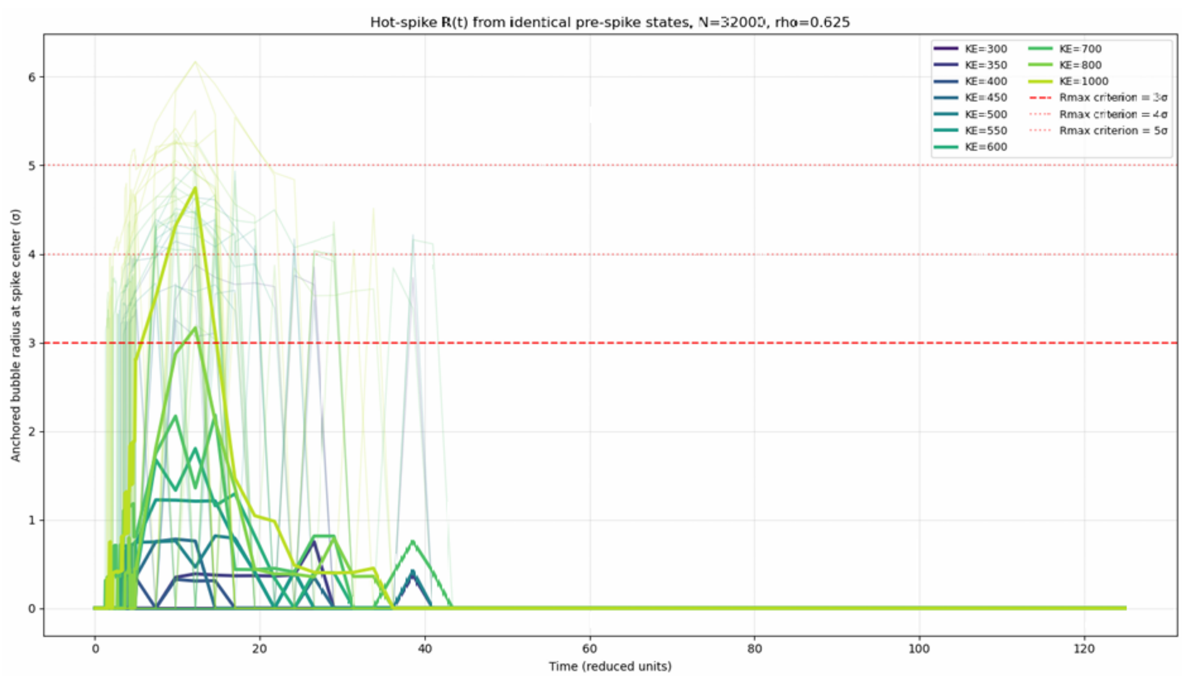

Figure 5.5: Bubble radius $R(t)$ from the spike center for all replicates at $N = 32{,}000$ and $\rho_l^* = 0.625$. The horizontal dashed lines indicate the $R_{\max} \geq 3\sigma$ (red), 4σ (blue), and 5σ (gray) nucleation criteria. At low energies, all curves remain below 3σ. Note that as the energy increases, there is an increase in the fraction of replicates producing transient bubbles that cross the threshold criteria.

At low KE injection energies ($KE \leq 400\,\varepsilon$), the bubble radius remains below $1\,\sigma$ for all replicates. As the energy increases through 450–600 ε, there is an increase in the fraction of replicates that produce transient bubbles whose radii briefly exceed 3σ before collapsing. At the high end of the KE injection scale ($KE \geq 800\,\varepsilon$), most replicates produce bubbles that reach 4–5 σ before the bubble collapses.

We show the corresponding $R(t)$ curves for the $N = 64{,}000$ in Figure 5.6.

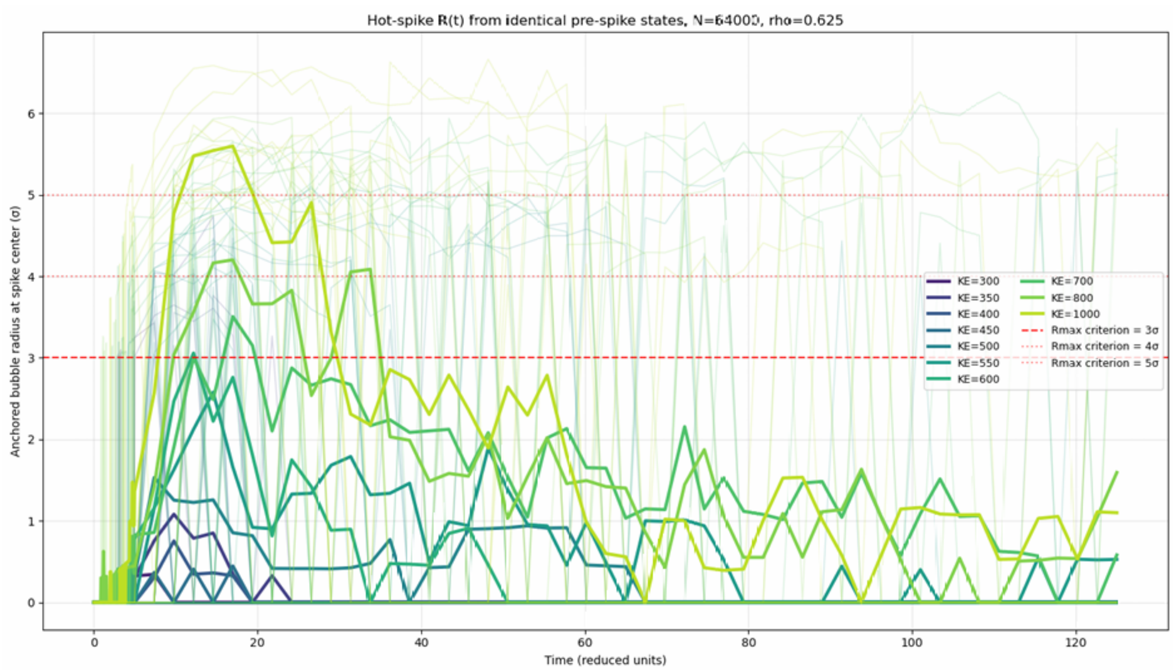

Figure 5.6: Bubble radius $R(t)$ from the spike center for all replicates at $N = 64{,}000$ and $\rho_l^* = 0.625$. The horizontal dashed lines indicate the $R_{\max} \geq 3\sigma$ (red), 4σ (blue), and 5σ (gray) nucleation criteria. The larger system size allows bubbles to reach larger maximum radii at the equivalent injected energies.

A particular characteristic feature of the $R(t)$ is the rapid initial growth followed by re-collapse. We reach the peak radius within 20–40 τ of the energy injection, after which the bubble shrinks and eventually disappears. The transient behavior encourages the use of $R_{\max}$, as the maximum radius captures the peak of the nucleation event, regardless of the collapse.

Chapter 6

Discussion

In this chapter, we bring together the seeding results and the hot-spike results and discuss the observed nucleation thresholds, compare the present simulations with previous studies, and assess the limitations.

6.1 Interpretation of the Hot-Spike Thresholds

The primary quantitative result of the hot-spike calculations is the dynamic threshold for transient bubble formation in a metastable Lennard-Jones liquid. Using the $R_{\max} \geq 4\sigma$ criterion. The threshold decreases from $600\,\varepsilon$ in the $N = 32{,}000$ system to $500\,\varepsilon$ in the $N = 64{,}000$ system. This indicates that we need much larger system sizes.

The hot-spike results show that the bubble formation in the present NVE simulations is transient. This justifies the use of $R_{\max}$ as the primary observable, as in this study, it captures whether the deposited energy is enough to produce a growing vapor cavity, although the box size does not allow the cavity to survive indefinitely.

We cannot determine a direct Seitz factor here. One would usually compare the dynamic hot-spike threshold with a thermodynamic threshold we derive from seeding. In the present work, the comparison is limited due to two issues. Firstly, the hot-spike simulations were performed at $\rho^* = 0.625$, while the seeding results were at $\rho_i^* = 0.61$. Secondly, the reported value of Q_{Seitz} depends on a reference-liquid choice that should more properly be matched to the equilibrated far-field liquid density $\rho_{l,f}$ of the seeded state. Given these reasons, we focus on the threshold itself rather than on a numerical Seitz factor.

6.2 Comparison with Literature

For comparing the hot-spike results, the most relevant study is the Denzel *et al.* [9]. They simulated bubble nucleation in an LJ fluid using elongated cylindrical energy deposition, compared to the spherical localized injection we used. Their simulations also contained about 20 million atoms per run and operated in a much larger size than our 32k and 63k systems. They found that the required total deposited energy was about twice the values expected from the Sietz picture, and stated this as a consequence of rapid thermal diffusion away from the deposition region. Although we cannot compare numerical Seitz factors directly, both studies illustrate the dynamic threshold and energy injection.

6.3 System-Size Effects and Limitations

The results from both system sizes reveal finite-size effects for the hot-spike simulation.

Threshold shift. For the the $R_{\max} \geq 4\sigma$ criterion, the threshold decreases by approximately $100\,\varepsilon$, from $600\,\varepsilon$ in the 32k system to $500\,\varepsilon$ in the 64k system. This indicates that the smaller system likely overestimates the dynamic threshold due to an increase in global temperature and the tendency for the bubble to re-collapse.

Maximum bubble radius. At fixed injected energy, the larger system also supports larger transient bubbles. For example, at $800\,\varepsilon$, $\langle R_{\max} \rangle$ increases from $4.46\,\sigma$ in the 32k system to $5.00\,\sigma$ in the 64k system, and at $1000\,\varepsilon$ it increases from $4.85\,\sigma$ to $5.84\,\sigma$. Therefore, the larger box permits more bubble growth before the finite-size effect suppresses it,

Late-time persistence. The late-time criterion never reaches the required 50% in either system, and only a small fraction of 64k runs exhibit nonzero late-time radii at the highest energies. This leads to the conclusion that even the 64k system is too small to support long-lived bubbles.

These trends show that finite-size effects do remain significant in our runs. Only a small fraction of 64k runs exhibit nonzero late-time radii at the highest energies. Therefore, for the next step, a reasonable increase in the system size should be by another factor of 4–8 in particle number, corresponding to $N \sim 2.5 \times 10^5$–5×10^5 particles. At $\rho^* = 0.625$, this would increase the box length to roughly $L \sim 74$–$93\,\sigma$, which would double the size of the 64k simulations and reduce the re-collapse seen in our finite NVE systems. At this proposed size, these runs would cost roughly 4–8× more compute time than the current 64k scans.

Therefore, finite-size effects combined with the difference in density in the seeding and hot-spike calculations are the limitations in our study.

6.4 Future Directions

Larger system sizes. It would be ideal if we could scale our simulations to those with $N \geq 10^5$–10^6 particles, which would reduce the finite-size effects, and we can observe a stable bubble growth.

Density Scans and Seitz factor Running the cavitation and the hot spike simulations at the same density will allow us to estimate the Setiz factor with much more accuracy, as the energy difference and the corrected term would both be calculated accordingly. The Seitz factor is extremely useful for input calibration for bubble chamber detectors; hence, a more refined and accurate measurement of f_{Seitz} as a function of deposition geometry and recoil energy will improve the theoretical predictions of the detector thresholds and might strengthen PICO-like experiments for future studies.

AI Use Statement. In preparing this thesis, I used Grammarly and ChatGPT for light editorial assistance, including spelling, punctuation, grammar, and limited wording revisions for clarity. No AI tools were used to generate figures.

Bibliography

[1] Gianfranco Bertone, Dan Hooper, and Joseph Silk. Particle dark matter: evidence, candidates and constraints. *Physics Reports*, 405(5):279–390, 2005. ISSN 0370-1573. doi: https://doi.org/10.1016/j.physrep.2004.08.031. URL `https://www.sciencedirect.com/science/article/pii/S0370157304003515`.

[2] C. Amole et al. Dark matter search results from the PICO-60 C_3F_8 bubble chamber. *Physical Review Letters*, 118:251301, 2017. doi: 10.1103/PhysRevLett.118.251301.

[3] D. Baxter, C. J. Chen, M. Crisler, et al. First demonstration of a scintillating xenon bubble chamber for detecting dark matter and coherent elastic neutrino-nucleus scattering. *Physical Review Letters*, 118:231301, 2017. doi: 10.1103/PhysRevLett.118.231301.

[4] P. Rosales-Pelaez, I. Sanchez-Burgos, C. Valeriani, C. Vega, and E. Sanz. Seeding approach to nucleation in the NVT ensemble: The case of bubble cavitation in overstretched Lennard-Jones fluids. *Physical Review E*, 101(2):022611, 2020. doi: 10.1103/PhysRevE.101.022611. Primary reference for NVT-Seeding methodology.

[5] Søren Toxvaerd and Jeppe C. Dyre. Communication: Shifted forces in molecular dynamics. *Journal of Chemical Physics*, 134(8):081102, 2011. doi: 10.1063/1.3558787. Truncated and force-shifted LJ potential methodology.

[6] M. P. Allen and D. J. Tildesley. *Computer Simulation of Liquids*. Oxford University Press, 2nd edition, 2017. ISBN 9780198803195.

[7] P. Rosales-Pelaez, M. I. Garcia-Cid, C. Valeriani, C. Vega, and E. Sanz. Seeding approach to bubble nucleation in superheated Lennard-Jones fluids. *Physical Review E*, 100(5): 052609, 2019. doi: 10.1103/PhysRevE.100.052609.

[8] Frederick Seitz. On the theory of the bubble chamber. *Physics of Fluids*, 1(1):2–13, 1958. doi: 10.1063/1.1724333. Original Seitz heat spike model.

[9] Philipp Denzel, Jürg Diemand, and Raymond Angélil. Molecular dynamics simulations of bubble nucleation in dark matter detectors. *Phys. Rev. E*, 93:013301, Jan 2016. doi: 10.1103/PhysRevE.93.013301. URL `https://link.aps.org/doi/10.1103/PhysRevE.93.013301`.